

CreditForYou: instant checkout credit for freelancers, gig workers & online sellers

Cash-flow rules decide today · ML learns from real repayments to sharpen limits tomorrow

1.8–4.6 days → instant

decision turnaround (case baseline, Exhibit 2)
→ pre-computed limit lookup, target < 1 s

380k → 50k VND

cost per decision: case benchmark
→ automated (team assumption)

16% → 51%

gig workers & online
merchants with a limit vs
proxy rule

simulated · n = 43

4.6%

break-even 6-month
default = stop-rule; RoC
28% in base case

simulated

① PROBLEM DIAGNOSIS

The checkout decision is welded to manual underwriting

No payslip · short or no bureau file · income arrives via platforms, e-wallets, online sales

Centralised manual review: documents, CIC check, credit officer

1.8–4.6 business days → the purchase moment is gone
Decline → lost sale for bank & merchant

The review costs more than a small loan earns: a 5M VND, 6-month loan at 12% earns ≈ **276k VND** (interest + 2% merchant fee) vs a **380k VND** review (Exhibit 2). Review only pays above ≈ **6.9M VND**, yet loans start at 1.6M.

Highest leverage: a faster review still costs 380k and keeps customers waiting; collections act only after a "yes". Taking the review out of checkout fixes speed *and* unit cost at once.

Root cause: income is real but invisible to a document review (stability, 1 – CV)



② SOLUTION CONCEPT · DATA-TO-DECISION ARCHITECTURE

Score cash-flow in advance; the till only checks the limit

1 · DATA INPUTS

What we read

- 12 months of CASA transactions
- Active loan payments
- Age, tenure, segment
- Basket value at checkout

Phase 2: platform / e-wallet data with consent. No payslip needed.

2 · LOGIC · NIGHTLY BATCH

How we decide

- Clean: drop own transfers, refunds, disbursements
- Income by source, gross→net (0.89 / 0.53)
- Capacity = min(50% disposable, 20% income, 50% income – debt)
- Tier A–D + guardrails
- Limit = min(capacity × annuity, caps)

3 · OUTPUTS

What we produce

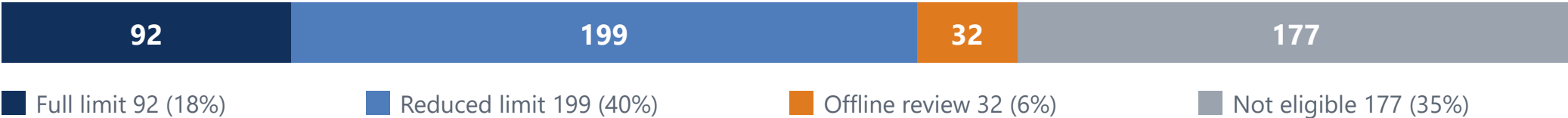
- Limit + status
- Reason codes
- Tier & APR band
- Audit trail

4 · BUSINESS ACTIONS

What the bank does

- Till:** real-time eKYC & fraud check, then approve / counter-offer / decline
- Targeting:** offers shown only to pre-approved customers
- 3 on-time → +25%; 30+ DPD → freeze
- ML:** repayment labels train a default model

Output on 500 simulated customers **SIMULATED** · median limit 15M VND · caps A 90M / B 60M / C 30M / new 30M · tenor 3–12 months



Guardrails only tighten: reject if debt service > 70%, spending > 120% or cash-flow deficit; limit ×0.5 if tier C, debt > 45% or spending > 90%. **Salaried with credit** (35.5% of mix) keep today's bureau terms via STP; cash-flow adds an affordability flag.

Community impact

Thin-file workers get a first formal credit line instead of informal credit; repaying grows the limit and a credit history.

Financially sustainable

Profitable while default < 4.6% (RoC 28%); stop-rule; rules in a config the credit committee owns.

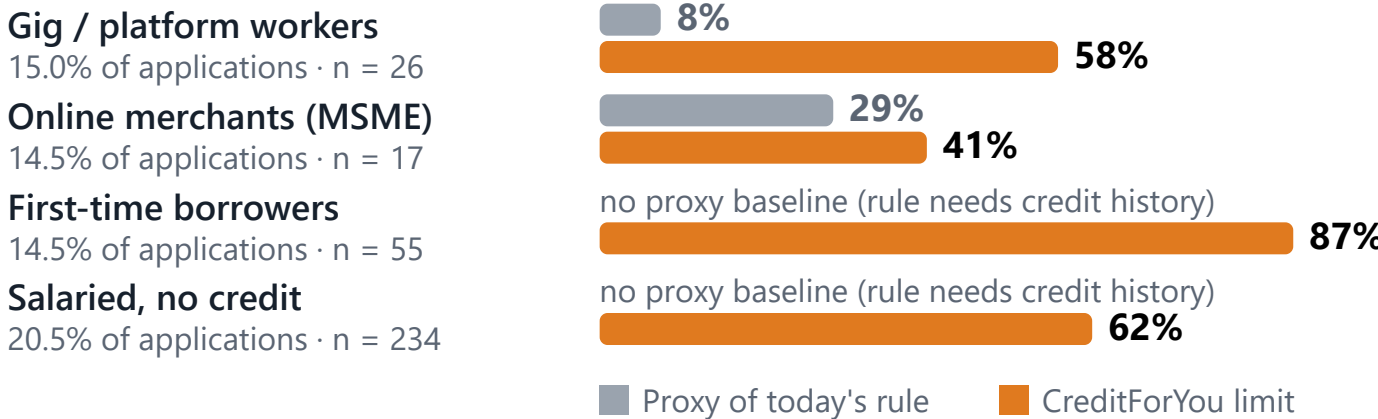
Environmentally light

Paperless, no documents to scan or courier; one nightly CPU batch on data the bank already holds.

Why ML is not the decision-maker yet: no POP default labels exist. A Lending Club PD model (300k loans, AUC 0.67) mis-ranks tiers (B 2.1% > C 1.9%), so it runs as a challenger until real labels beat the rules.

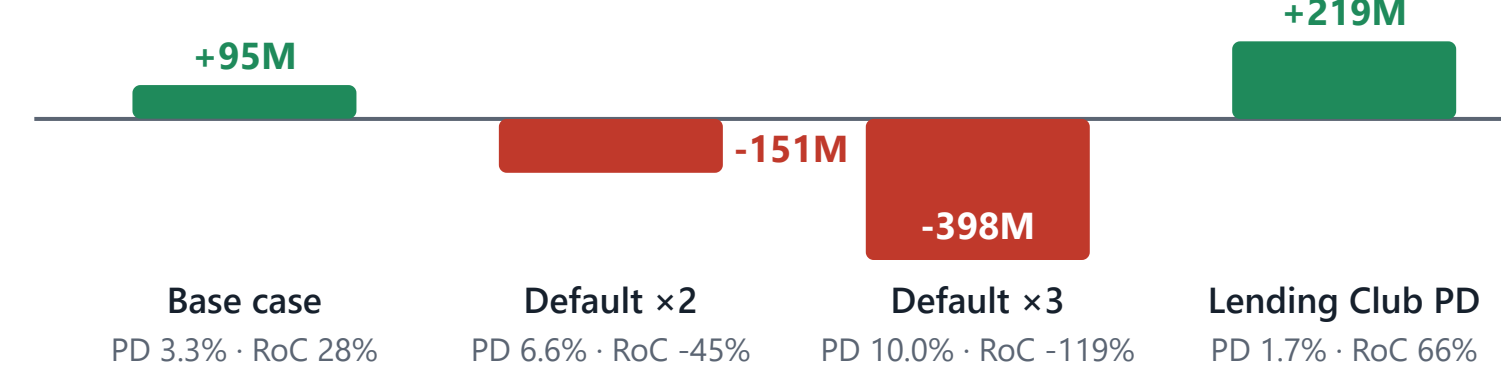
③ INTERVENTION JUSTIFICATION · WHY IT WORKS AND PAYS

3× more gig workers & online merchants served, profitable while default < 4.6%



Reweight to the Exhibit 2 mix: instant limits for **56%** of inbound applications. Gig + MSME n = 43 (95% CI 37%–65%); re-validate in the backtest.

Annual profit, million VND, by default scenario (291 simulated customers with a limit)



- Speed & cost:** 1.8–4.6 days, 380k/file → instant, ≈ 50k; 199 files skip review.
- Reach:** +211 newly served, –93 over-indebted (mortgage-heavy) declined → net +118.
- Value:** profit 95M VND/yr, RoC 28%; +1 pp merchant fee = +113M VND.

Downside is real, and capped: at ×2 / ×3 default, losses reach 4.3% / 6.5% of volume vs break-even 3.0% (US BNPL charge-off 3.8%). Controls: stop-rule at 4.6% PD, 30M VND starter cap, ×0.5 reduced limits, short tenor, fixed pilot loss budget, pricing lever.

④ FEASIBILITY · 6–12 MONTHS, EXISTING CUSTOMERS FIRST

0–3 mo · Backtest

Run read-only on real HLBVN accounts; re-learn income factors; committee signs off the config.

Gate: income error ≤ 10%; segment results hold at full sample

3–6 mo · Pilot

1–2 merchants, low caps, fixed loss budget, weekly arrears tracking.

Gate: 6-mo default ≤ 3%; 30+ DPD A < 2%, B < 5%

6–12 mo · Scale + ML

More merchants, limit growth on, partner data, ML challenger on repayment labels.

Stop: default > 4.6% (break-even)

⑤ SOLUTION CONCEPT · KPIS MEASURING BOTTLENECK RELIEF

KPI	Baseline (case)	Target	Sim.
Decision turnaround	1.8–4.6 days	< 1 s	lookup
Cost per decision	380k VND	≤ 50k VND	assumed
Checkout conversion*	n/a	≥ 60%	58%
Gig & MSME with a limit	16% proxy	≥ 50%	51%
Files sent to offline review	100% manual	≤ 10%	6.4%
6-month default (stop > 4.6%)	n/a	≤ 3%	pilot

* approve + counter-offer on a 5M VND basket

⑥ DATA & ASSUMPTIONS

Illustrative simulation: stand-in synthetic data (500 customers, 1.67M transactions, USD → VND at 25,000), reweighted to the Exhibit 2 mix; official case data arrives in the Strategy round. Cleaned in code: 7,730 mislabelled salary credits, 31,324 own transfers, 133 mortgages out of model DTI.

Manual review 380k VND · 1.8–4.6 days · 1.6–90.1M VND · 3–24 mo	Case, Exhibit 2
LGD 91% · EAD 72%	Lending Club, 90k defaulted loans
Base PD A 2% / B 5% / C 10%	≈ 2× Lending Club grades A–C (6-mo)
Merchant fee 2% · funding 6%	below CFPB BNPL 2.49% · above VN deposits 4.6–5.8%
Automated decision 50k VND	team assumption, to confirm with HLBVN